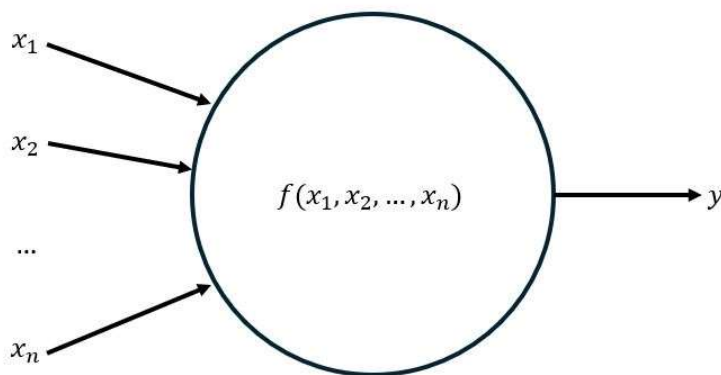
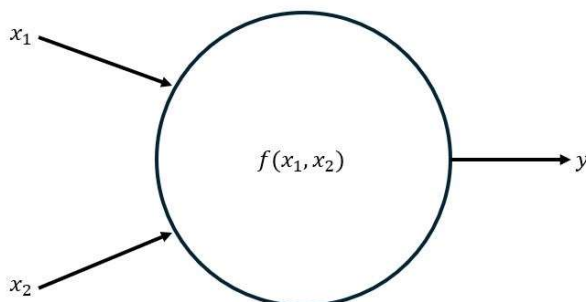


1. Receive input from neighbours.
2. Compute whether ^{total} signal exceeds threshold.
3. Produce output (if total signal exceeds threshold).
4. Connect output to other neurons with some strength.



Example
AND Neuron



Truth table

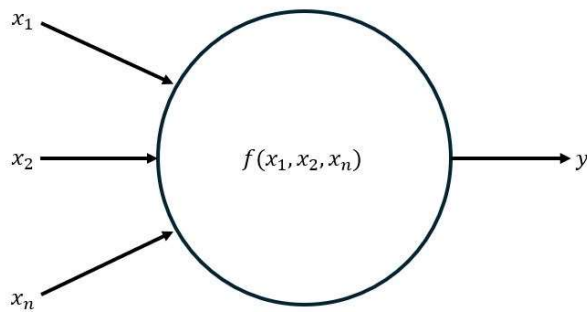
x_1	x_2	$\text{AND}(x_1, x_2)$
0	0	0
0	1	0
1	0	0
1	1	1

x_1, x_2 are excitatory

$$f(x_1, x_2) = x_1 + x_2$$

$$y = \begin{cases} 1 & \text{if } f(x_1, x_2) \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

Example OR Neuron



Truth table

x_1	x_2	x_3	OR
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

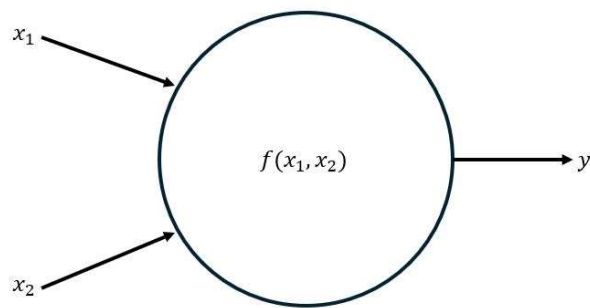
$$F(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

$$y = \begin{cases} 1 & \text{if } F(x_1, x_2, x_3) \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

Example

OR(x_1 , NOT(x_2)) Neuron

Truth table



x_1	x_2	output
0	0	1
0	1	0
1	0	1
1	1	1

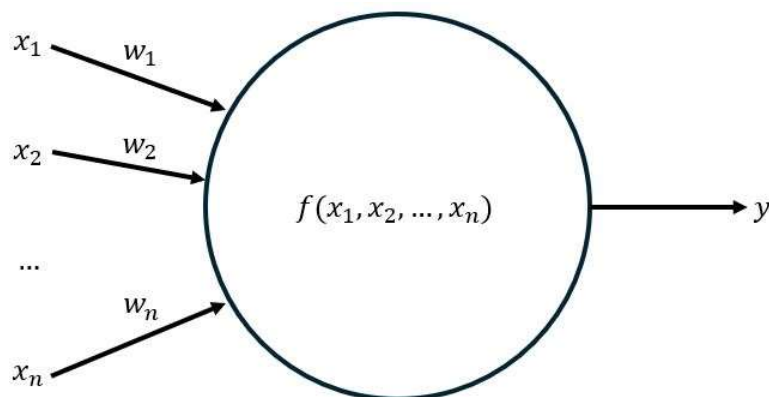
$$f(x_1, x_2) = x_1 - x_2$$

$$y = \begin{cases} 1 & \text{if } f(x_1, x_2) \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Perceptron Model

Connections have real-valued weights

$$f(x_1, \dots, x_n) = \sum_{i=1}^n w_i \cdot x_i$$



$$y = \begin{cases} 1 & \text{if } f(x_1, \dots, x_n) \geq \text{threshold} \\ 0 & \text{otherwise} \end{cases}$$

"binary"

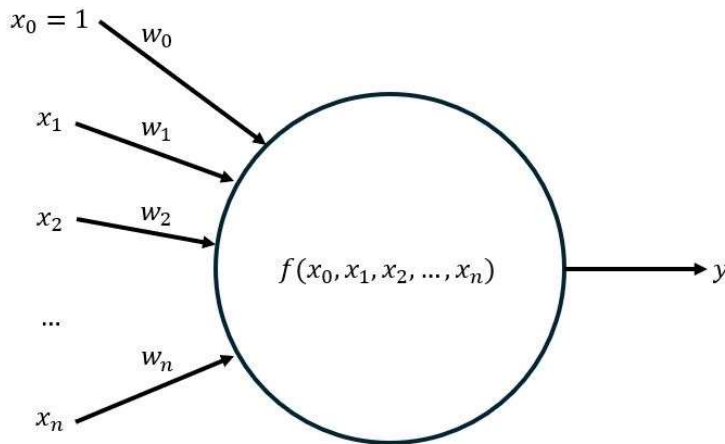
$$y = \begin{cases} 1 & \text{if } f(x_1, \dots, x_n) \geq \text{threshold} \\ -1 & \text{otherwise} \end{cases}$$

"bipolar"

Add a constant input $x_0 = 1$
with associated weight w_0 .
"bias term"

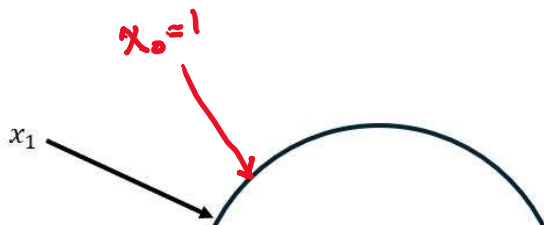
$$f(x_0, x_1, \dots, x_n) = \sum_{i=0}^n w_i \cdot x_i$$

$$y = \begin{cases} 1 & \text{if } f(x_0, \dots, x_n) \geq 0 \\ 0 & \text{otherwise} \end{cases}$$



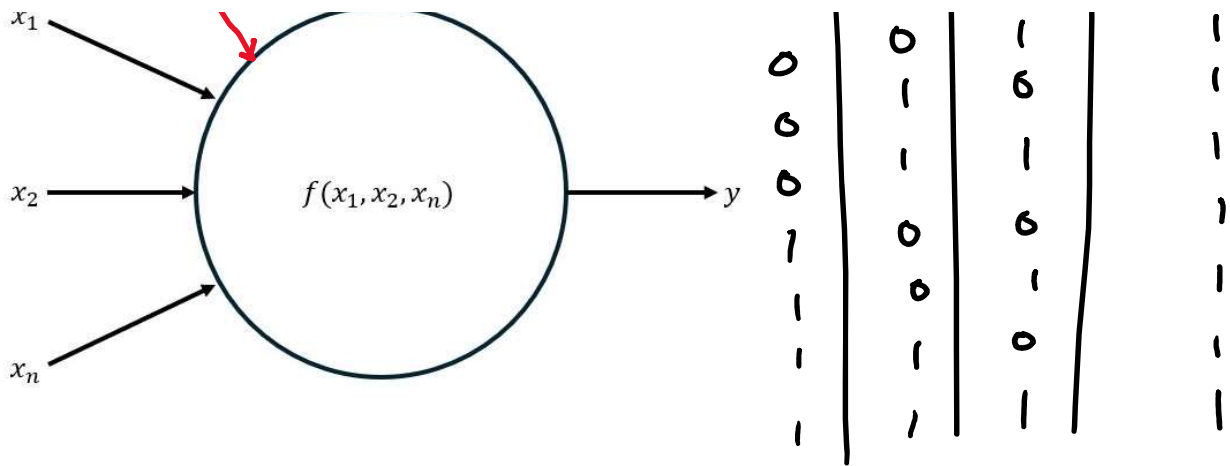
Example

OR Neuron



Truth table

x_1	x_2	x_3	OR
0	0	0	0
0	0	1	1
0	1	0	1



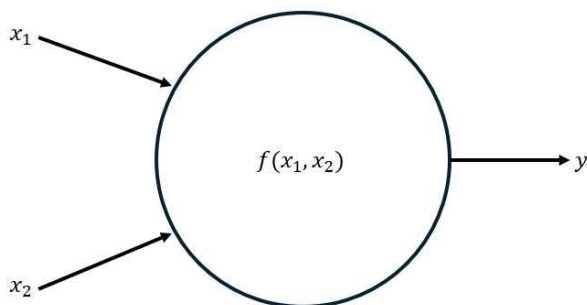
$$f(x_0, x_1, x_2, x_3) = w_0 x_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$y = \begin{cases} 1 & \text{if } f(x_0, \dots, x_3) \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$w_0 = -1 \quad w_1 = w_2 = w_3 = 1$$

Example

XOR(x_1, x_2)



x_1	x_2	XOR
0	0	0
0	1	1
1	0	1
1	1	0

$$F(x_0, x_1, x_2) = w_0 x_0 + w_1 x_1 + w_2 x_2$$

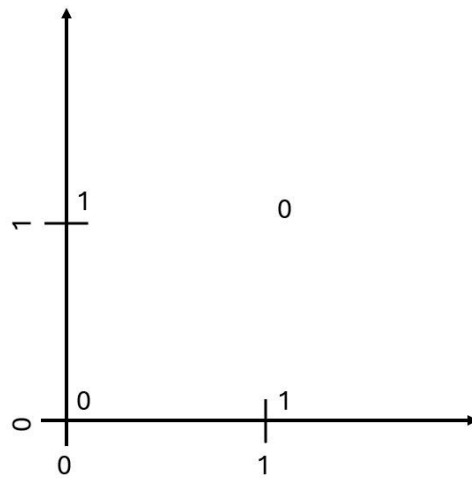
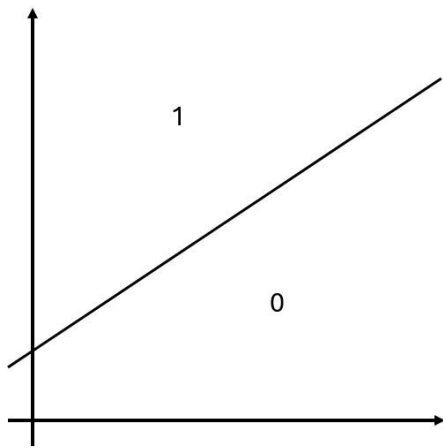
$$y = \begin{cases} 1 & F(x_0, x_1, x_2) \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$y = \begin{cases} 1 & \text{otherwise} \end{cases}$$

$$w_0 = 2, \quad w_1 = w_2 = -1$$

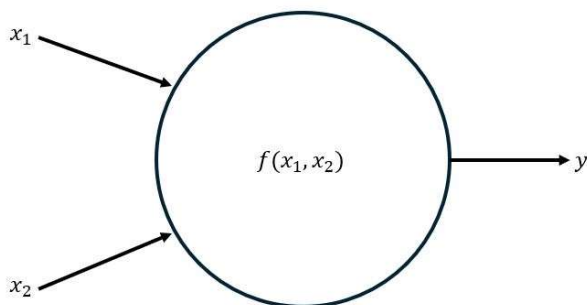
$$F(x_1, x_2) = w_0 + w_1 x_1 + w_2 x_2$$

$$0 = w_0 + w_1 x_1 + w_2 x_2$$



Example

NAND Neuron



x_1	x_2	NAND
0	0	1
0	1	1
1	0	1
1	1	0

$$f(x_0, x_1, x_2) = w_0 + w_1 x_1 + w_2 x_2$$

$$y = \begin{cases} 1 & \text{if } f(x_0, x_1, x_2) \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$w_0 = 1, \quad w_1 = -1, \quad w_2 = -1$$

Property:

We can write any boolean function as a network of NAND neurons.

In general, we can write any function as a network of neurons.

Aggregation + Activation Functions

Aggregation function:

$$f(x_0, \dots, x_n) = \sum_{i=0}^n w_i \cdot x_i$$

Activation function:

1. $a(\cdot)$ is non-decreasing

2. $a(\cdot)$ is non-linear.

1. $a(\cdot)$ is non-linear.
2. $a(\cdot)$ is non-linear.

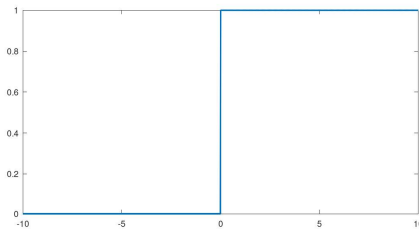
Heaviside Activation

$$a(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

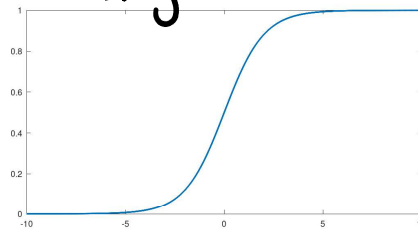
Logistic Function

$$a(x) = \frac{1}{1 + e^{-x}}$$

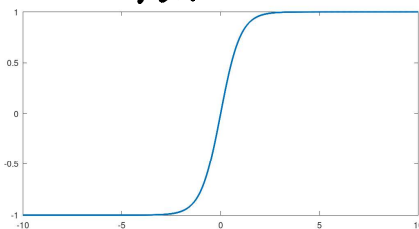
Heaviside



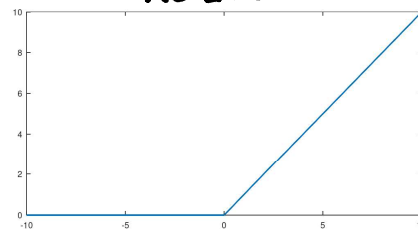
Logistic



tanh



ReLU



Tanh Activation

$$a(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Rectified Linear Unit (ReLU) Activation

$$a(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$